

COMP9418: Advanced Topics in Statistical Machine Learning

Bayesian Networks

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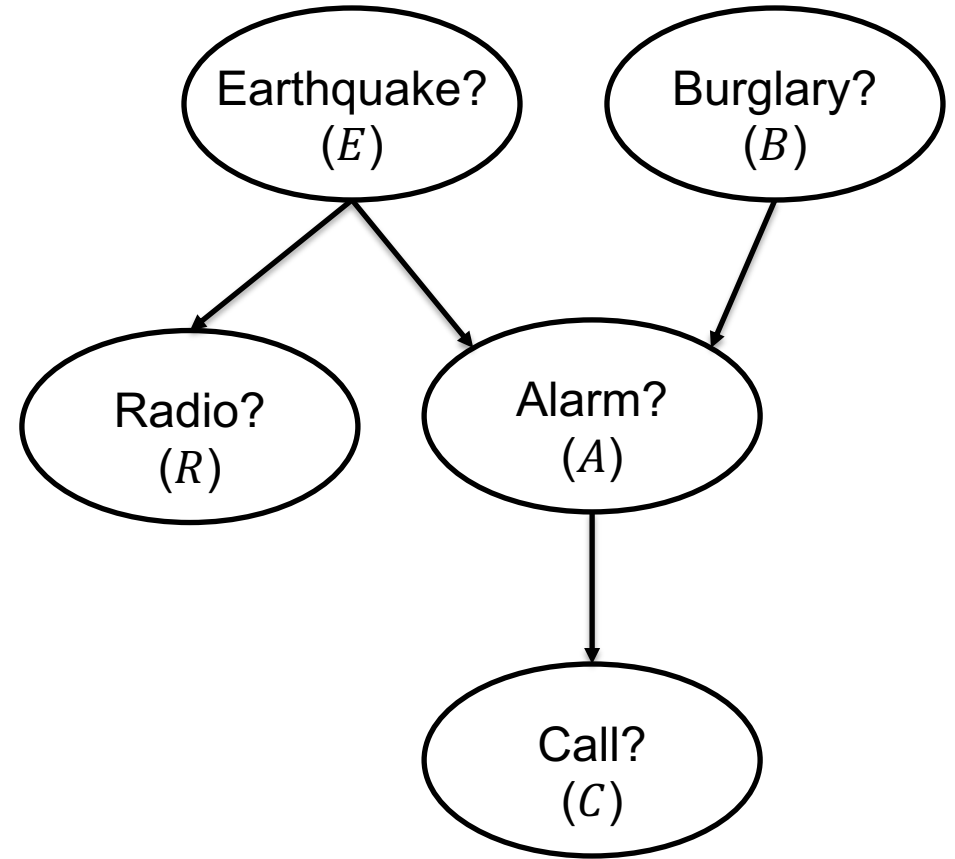
University of New South Wales

Introduction

- This lecture introduces Bayesian networks as a modelling tool to specify joint probability distributions
 - The size of a joint distribution is exponential in the number of variables
 - This causes modelling and computational difficulties
 - The specification of a joint distribution may hide some relevant properties such as independencies
- Bayesian networks is a graphical modelling tool for specifying probability distributions
 - It relies on the insight that independence is a significant aspect of beliefs, and
 - Independencies can be elicited using the language of graphs

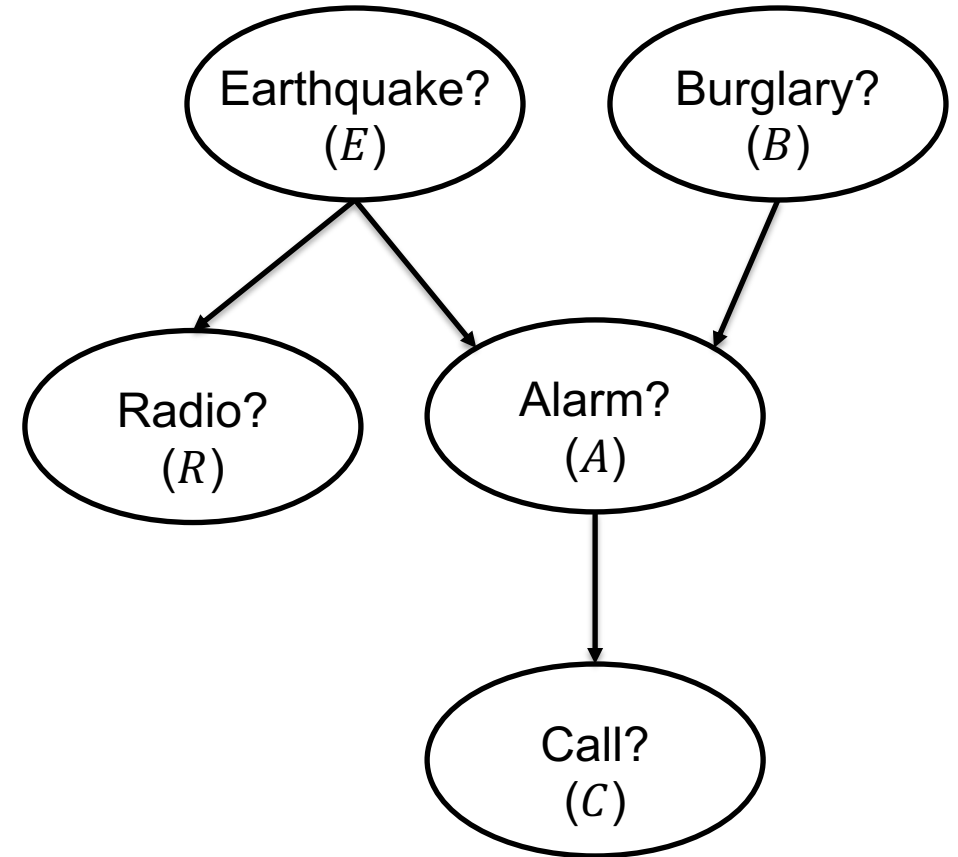
Graphs and Independence

- This figure is a *directed acyclic graph (DAG)*
 - Nodes represent variables
 - Let us assume (for now) that edges represent “direct causal influence”
 - For example, alarm triggering (A) causes a call for a neighbour (C)
- Given this representation, we expect the belief dynamic to satisfy some properties
 - For instance, C is influenced by evidence on R
 - A radio report would increase belief in Alarm. In turn, increase belief in a call from a neighbour
 - However, the belief in C would not increase if we knew the alarm did not trigger
 - $C \perp R | \neg A$



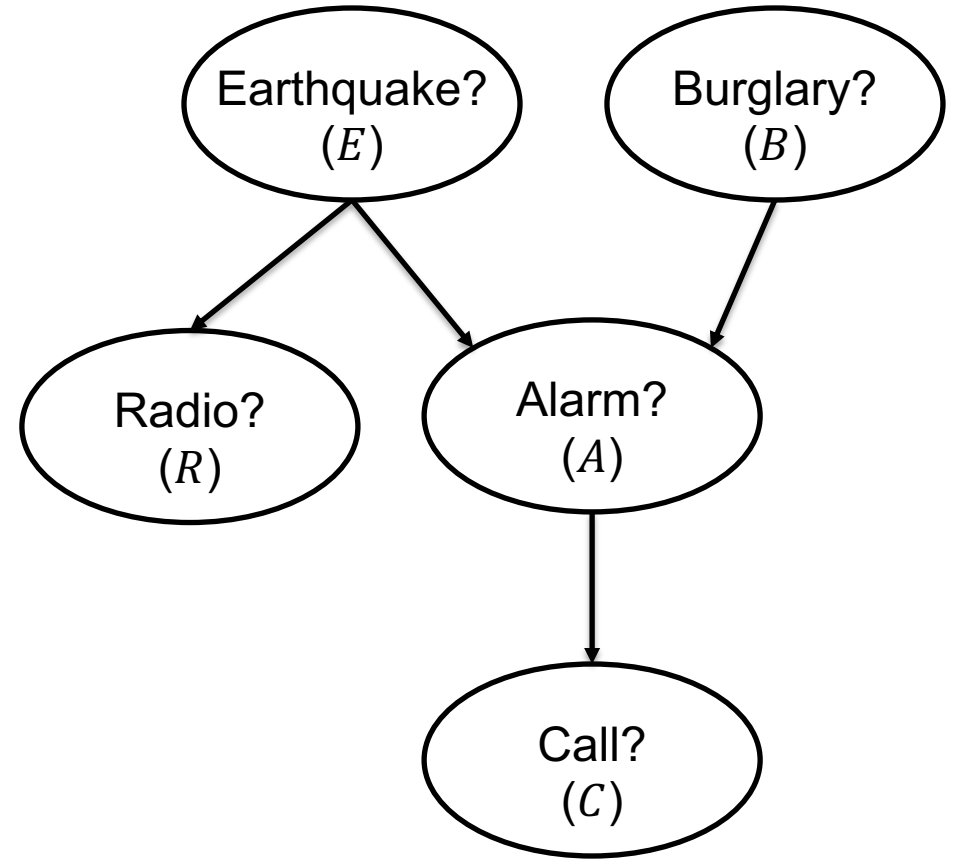
Notation

- Given a variable V in a DAG G
 - $Parents(V)$ are the parents of V in DAG G , that is, the set of variables N with an edge from N to V
 - $Descendants(V)$ are the descendants of V in G , that is, the set of variables N with a direct path from V to N
 - $Non_Descendants(V)$ are all variables in G other than V , $Parents(V)$, and $Descendants(V)$
- A DAG G is a compact representation of the following independence statements
 - $V \perp Non_Descendants(V) \mid Parents(V)$
 - Every variable is conditionally independent of its nondescendants given its parents
 - *Markovian assumptions* of G denoted by $Markov(G)$



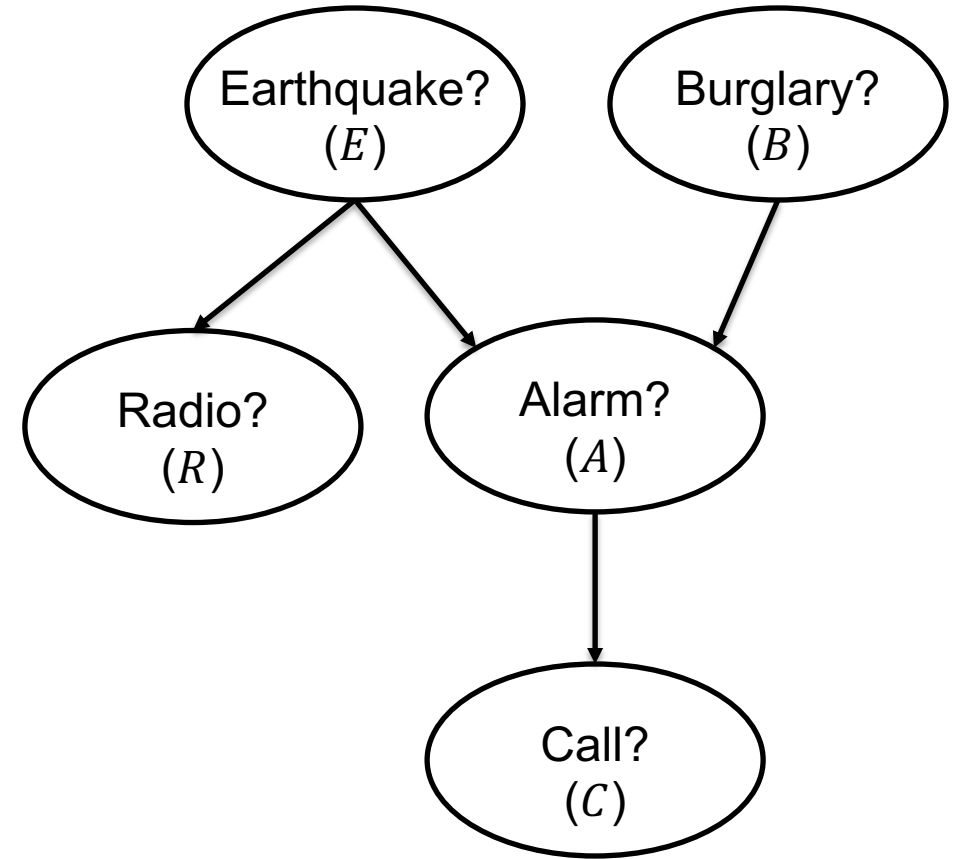
Markovian Assumptions

- If we view DAG G as a causal structure
 - $\text{Parents}(V)$ are direct causes of V
 - $\text{Descendants}(V)$ denotes the effects of V
- Given the direct causes of a variable, our beliefs in that variable will no longer be influenced by any other variable except possibly by its effects
- These are all the statements in this DAG



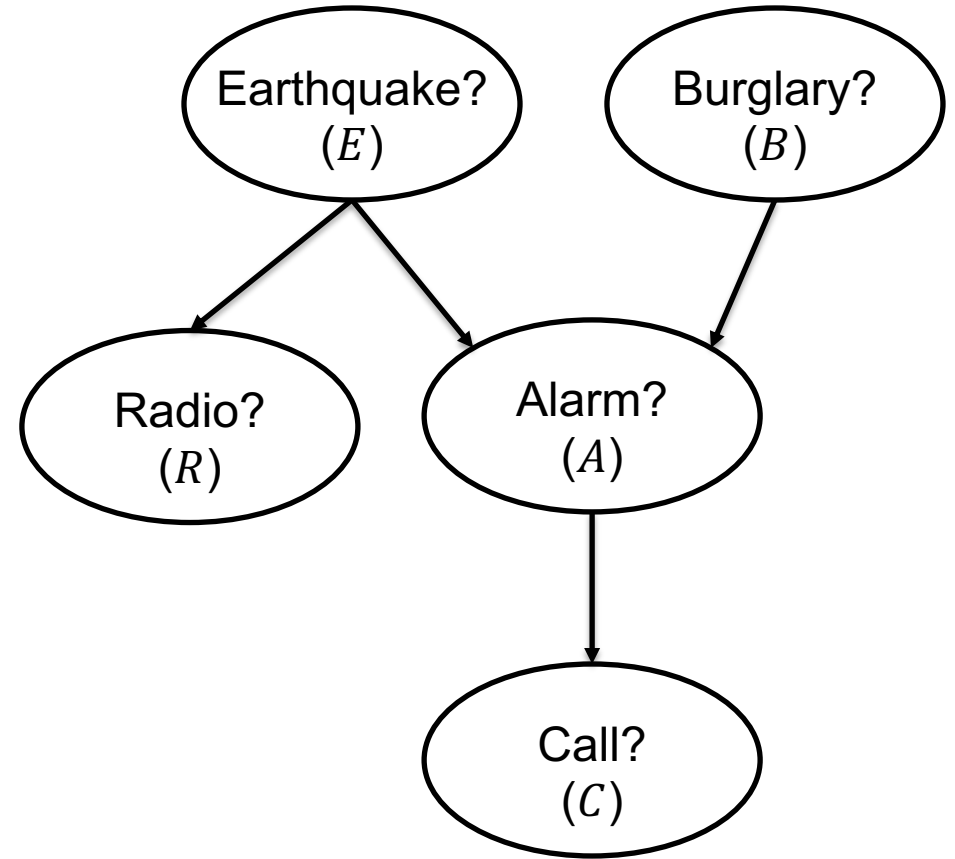
Markov Assumptions

- If we view DAG G as a causal structure
 - Parents(V) are direct causes of V
 - Descendants(V) denotes the effects of V
- Given the direct causes of a variable, our beliefs in that variable will no longer be influenced by any other variable except possibly by its effects
- These are all the statements in this DAG
 - $C \perp B, E, R \mid A$
 - $R \perp A, B, C \mid E$
 - $A \perp R \mid B, E$
 - $B \perp E, R$
 - $E \perp B$



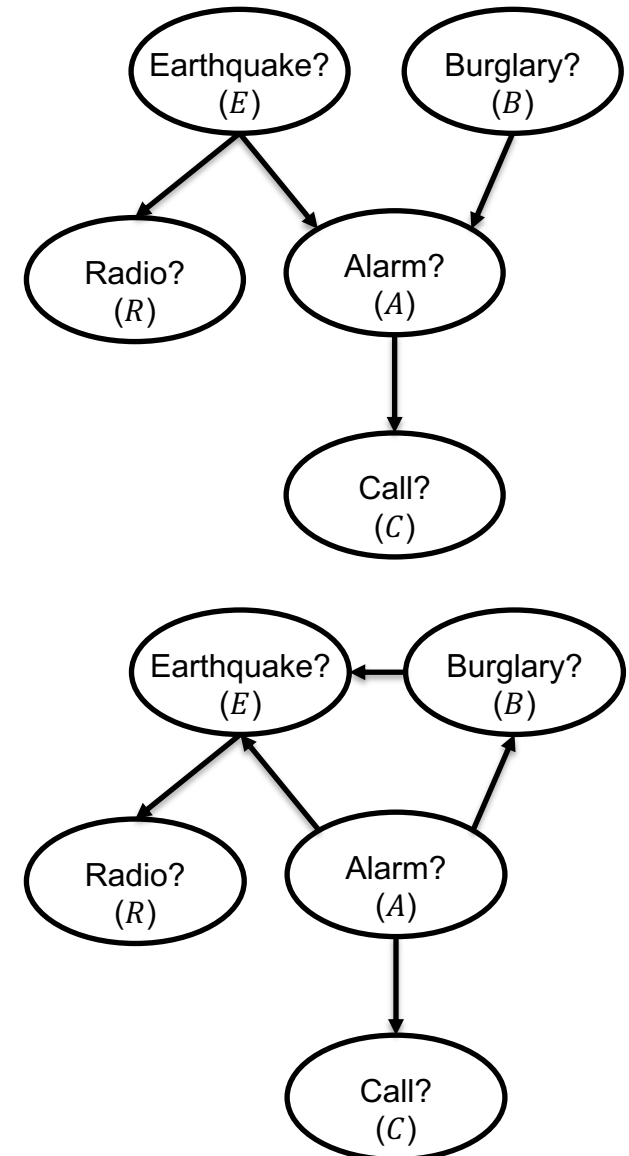
Markov Assumptions

- Suppose we want to make a probability distribution that captures the state of belief
 - The first step is to construct the graph, ensuring the independences on G matches our beliefs
 - The DAG G is a partial specification. It says that P must satisfy $\text{Markov}(G)$
- The specification of G restricts the choices for the distribution P
 - However, it does not uniquely define it
 - We need to augment G with a set of conditional probabilities
 - The conditional probabilities and G are guaranteed to uniquely define the distribution P



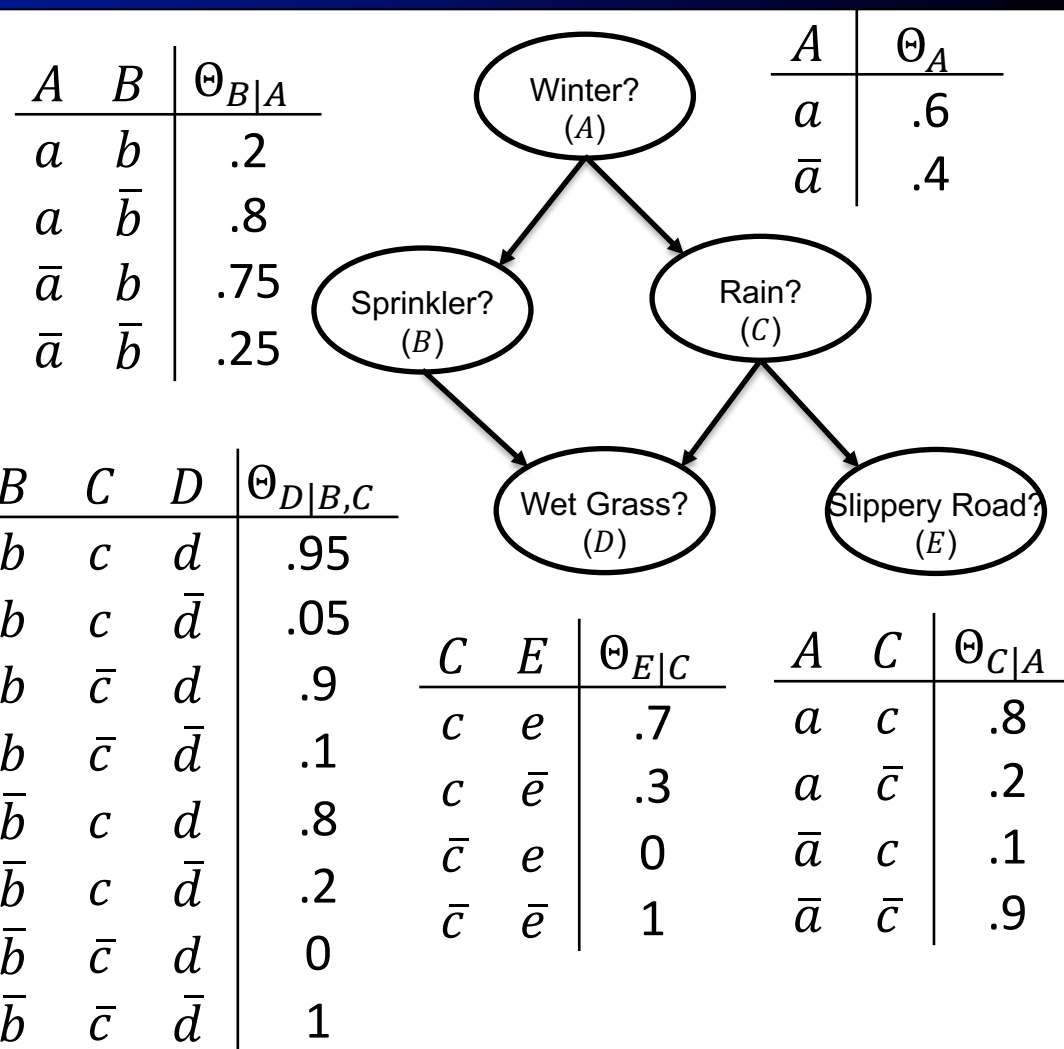
Causality

- The formal interpretation of a DAG is a set of conditional independences
 - It makes no reference to causality
 - However, we used causality to motivate this interpretation
- It is perfectly possible to have a DAG that does not match our causal perception
 - We will see that every independence in the first graph is also present in the second
 - We discuss next the graph parametrization (quantifying dependencies between nodes and parents)
 - This process is much easier to accomplish by an expert if the DAG corresponds to causal perceptions



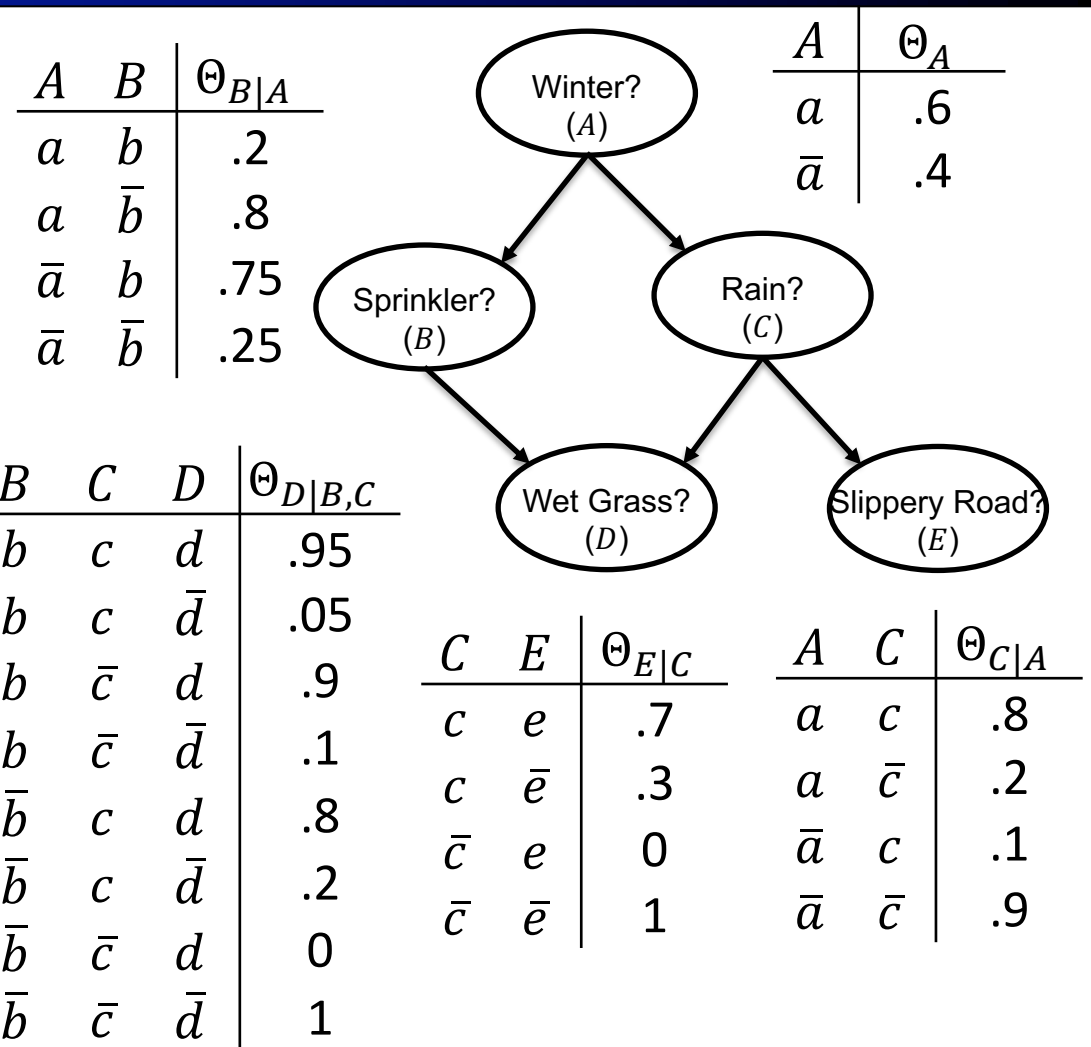
Parametrisation

- The conditional probabilities we need to specify are
 - For every variable X in DAG G and its parents $\mathbf{U}$
 - Provide the probabilities $P(x|\mathbf{u})$ for every value x of X and every instantiation $\mathbf{u}$ of parents $\mathbf{U}$
- For example, for this graph, we need to specify
 - $P(B|A), P(E|C), P(C|A), P(A), P(D|B, C)$
 - Each table is known as a *conditional probability table* (CPT)
 - Notice that $\sum_x P(x|\mathbf{u}) = 1$ for each $\mathbf{u} \in \mathbf{U}$
- Therefore, we only need 11 probabilities to specify the CPTs of this graph



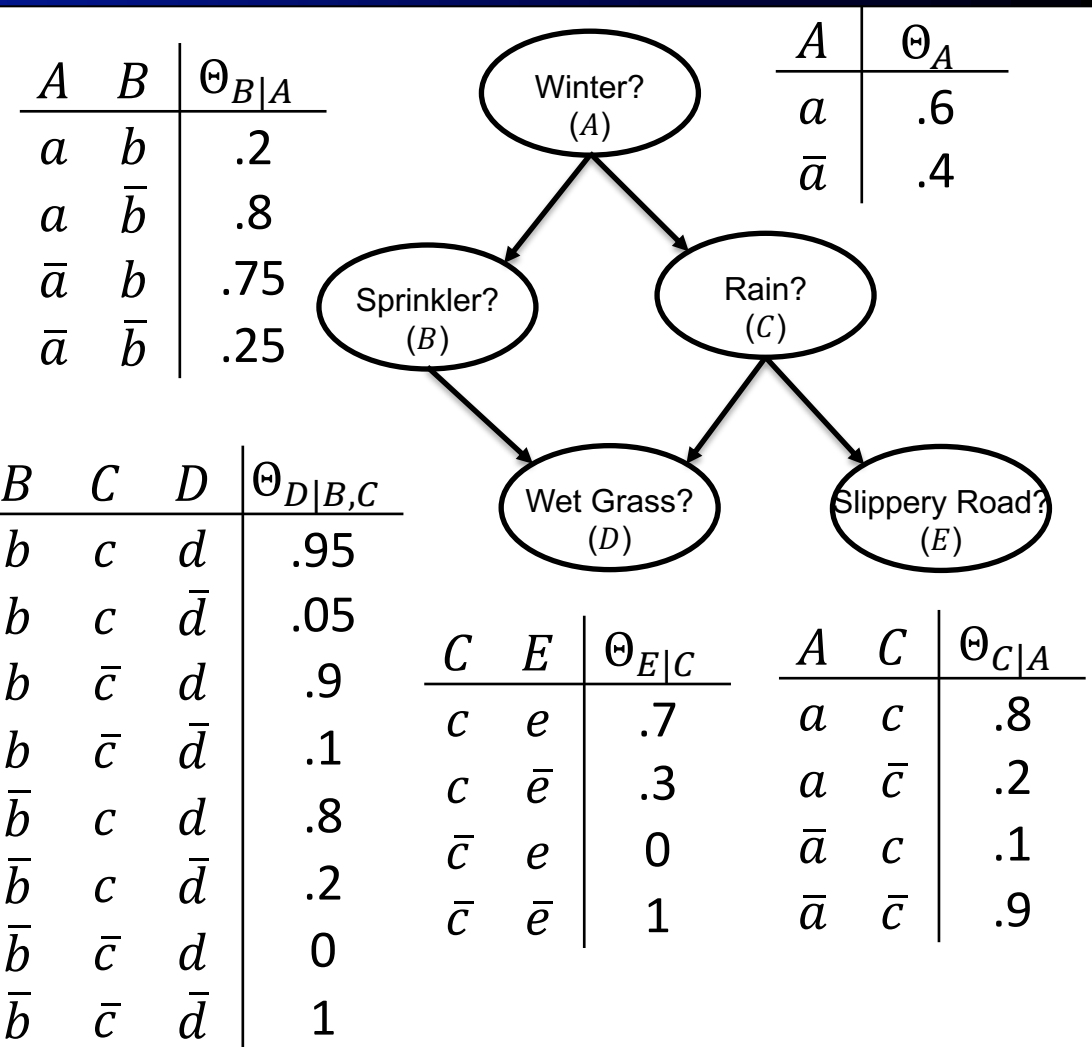
Bayesian Networks: Definition

- A Bayesian network for variables $\mathbf{Z}$ is a pair (G, Θ) , where
 - G is a directed acyclic graph over variables $\mathbf{Z}$, called the *network structure*
 - Θ is a set of CPTs, one for each variable in $\mathbf{Z}$, called the *network parametrization*
- We use
 - $\Theta_{X|U}$ to denote the CPT for variable X and its parents U
 - XU to denote a set of variables known as *network family*
 - $\theta_{x|u}$ is the value of $P(x|u)$ known as *network parameter*



Bayesian Networks: More Definition

- *Network instantiation* is an assignment of all network variables
 - A network parameter $\theta_{x|\mathbf{u}}$ is compatible with a network instantiation $\mathbf{z}$ when $x\mathbf{u}$ and $\mathbf{z}$ agree on common variables
 - We write $\theta_{x|\mathbf{u}} \sim \mathbf{z}$
 - For instance, θ_a , $\theta_{b|a}$, $\theta_{\bar{c}|a}$, $\theta_{d|b,\bar{c}}$, and $\theta_{\bar{e}|\bar{c}}$ are parameters compatible with the instantiation $a, b, \bar{c}, d, \bar{e}$.



Bayesian Networks: More Definition

- Only one probability distribution satisfies the constraints imposed by a Bayesian network

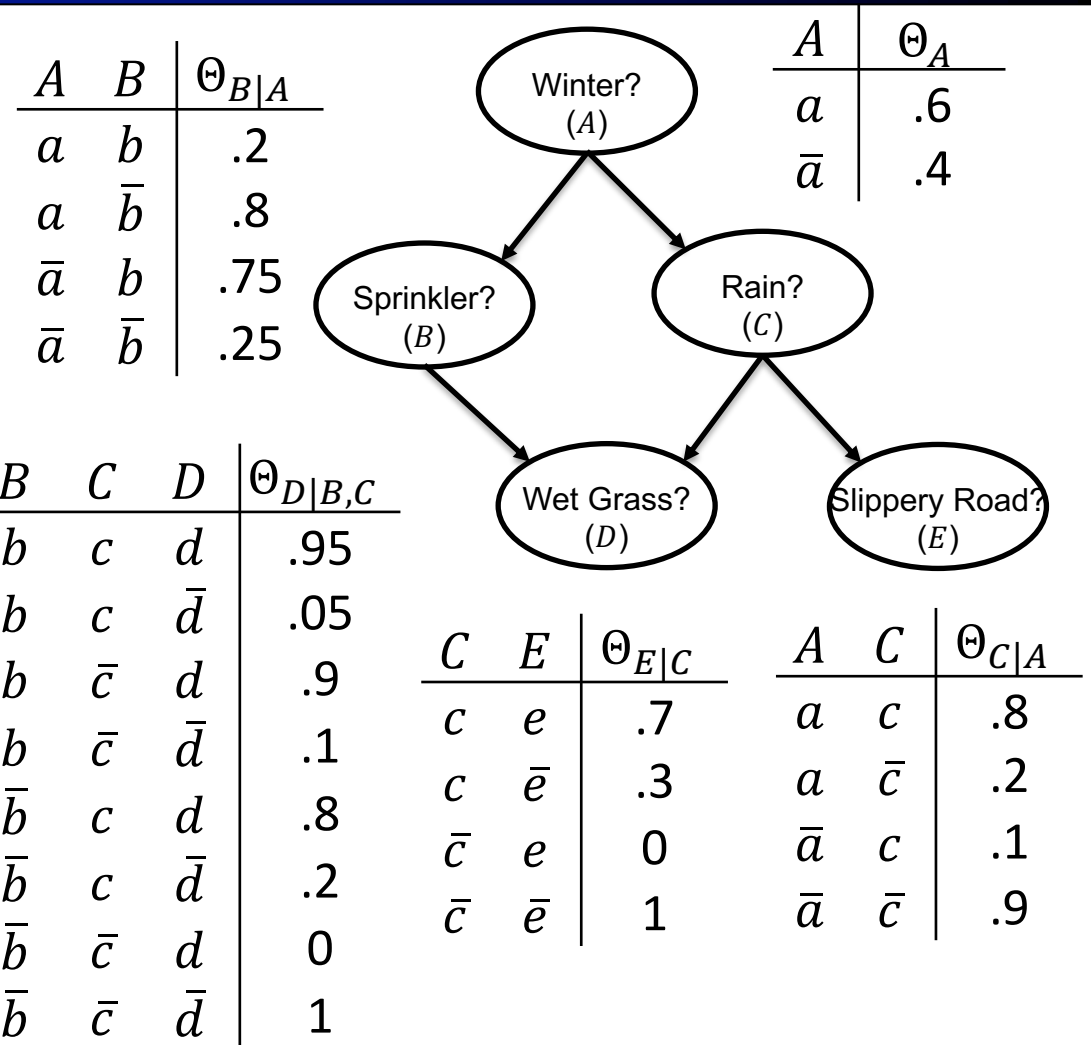
- The distribution is given by

$$P(\mathbf{z}) \stackrel{\text{def}}{=} \prod_{\theta_{x|u} \sim \mathbf{z}} \theta_{x|u}$$

- This equation is known as the *chain rule* for Bayesian networks

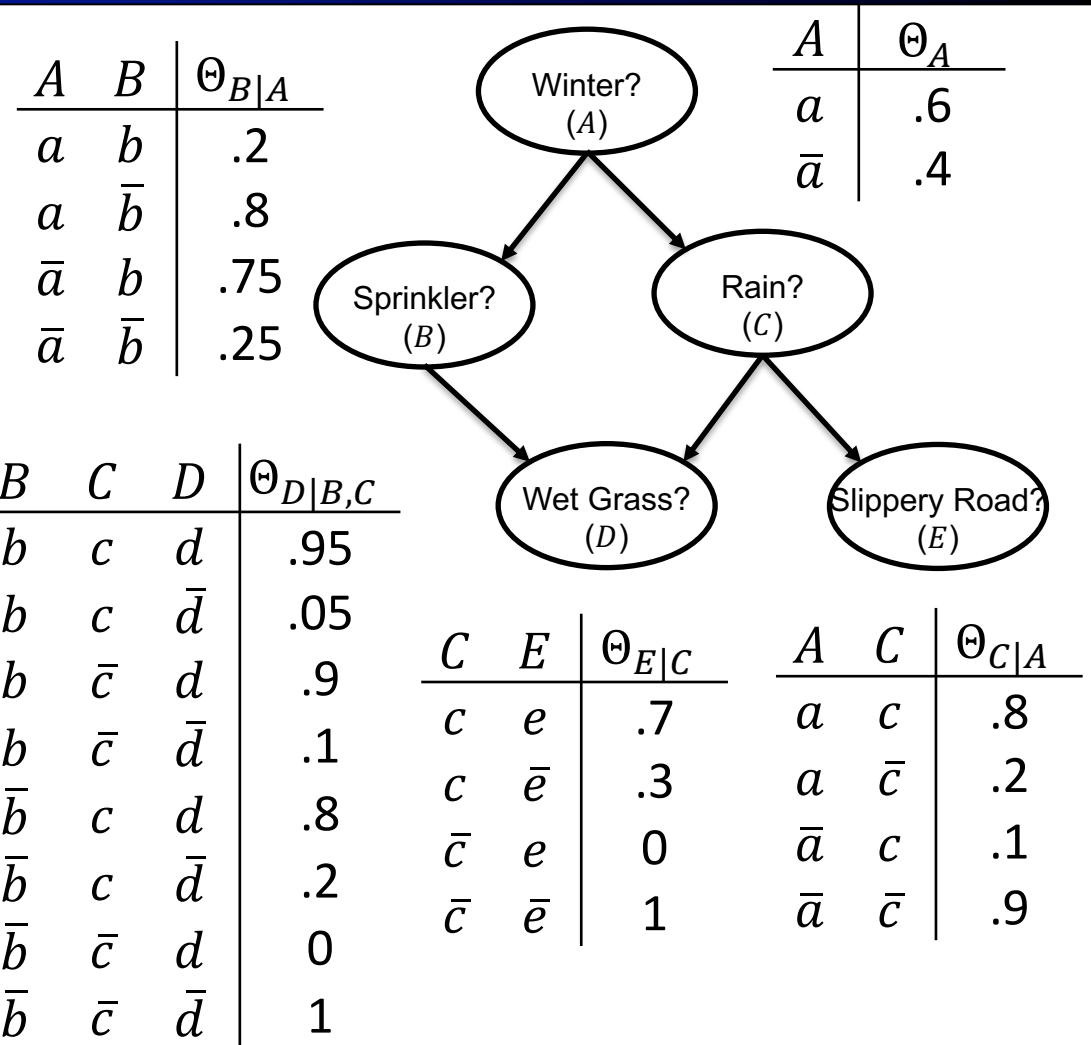
- For instance,

$$\begin{aligned} P(a, b, \bar{c}, d, \bar{e}) &= \theta_a \theta_{b|a} \theta_{\bar{c}|a} \theta_{d|b, \bar{c}} \theta_{\bar{e}|\bar{c}} \\ &= (.6)(.2)(.2)(.9)(1) \\ &= .0216 \end{aligned}$$



Bayesian Networks: Complexity

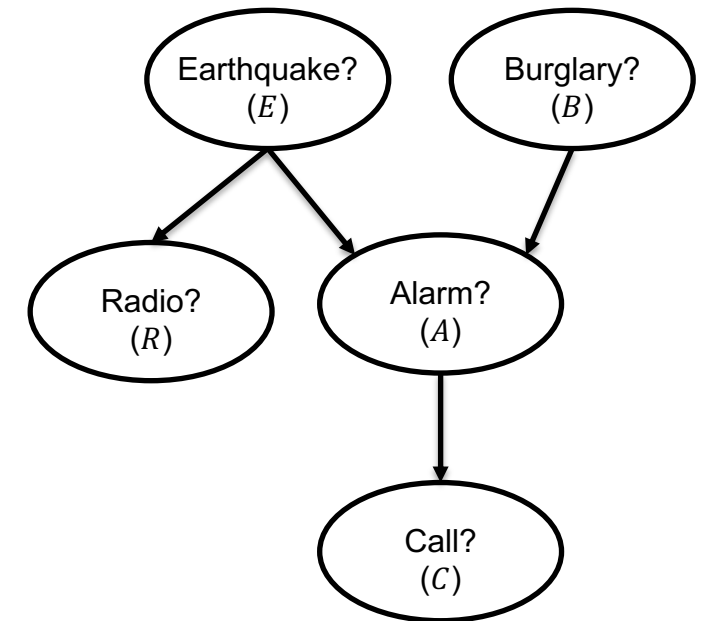
- The size of the CPT $\Theta_{X|U}$ is exponential in the number of parents U
 - If the maximal number of parents for every variable is k then the size of any CPT is $O(d^{k+1})$, where d is the number of values
 - With n network variables, the total number of variables is bounded by $O(nd^{k+1})$
- This number is reasonable if the number of parents is small
 - We will discuss techniques to represent CPTs when the number of parents is large



Properties of Independence

- The distribution P specified by a Bayesian network (G, Θ) satisfies the independence assumptions in $\text{Markov}(G)$
 - However, these are not the only independences satisfied by P
 - For example, $R \perp A \mid E$
- This independence and other ones follow the ones in $\text{Markov}(G)$
 - If we use a set of properties known as *graphoid axioms*
 - These properties include symmetry, decomposition, weak union and contraction

$$X \perp \text{Non_Descendants}(X) \mid \text{Parents}(X)$$



Properties of Independence: Symmetry

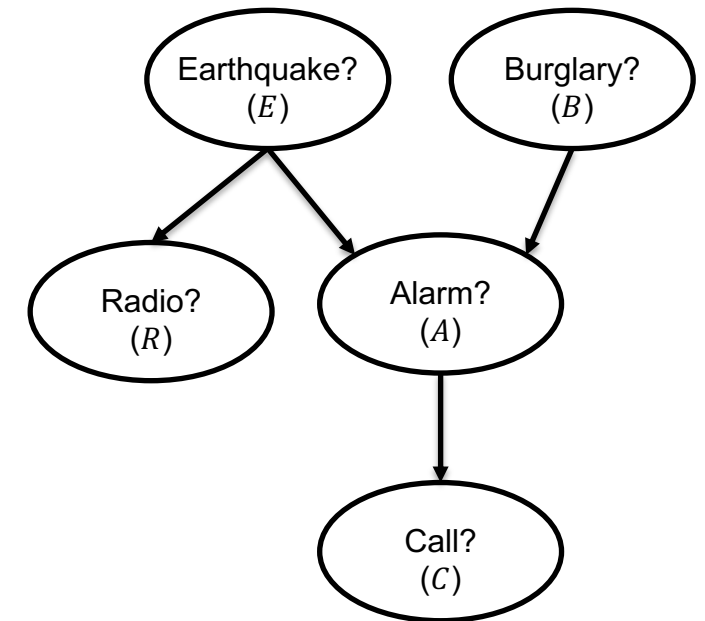
- *Symmetry* is the simplest property of probabilistic independence

- If learning y does not influence our belief in x , then learning x does not influence our belief in y .

- In the example graph

- $A \perp R \mid B, E$ (Markovian property for A)
- $R \perp A \mid B, E$ (using symmetry)

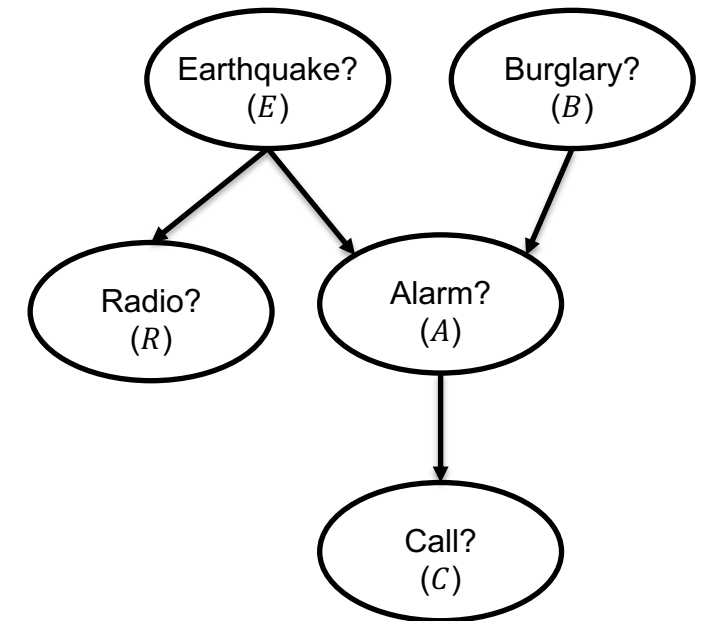
$X \perp Y \mid Z$ if and only if $Y \perp X \mid Z$



Properties of Independence: Decomposition

- The second property is *decomposition*
 - If learning $\mathbf{y}\mathbf{w}$ does not influence our belief in $\mathbf{x}$, then learning $\mathbf{y}$ alone, or learning $\mathbf{w}$ alone, does not influence our belief in $\mathbf{y}$.
- In the example graph
 - $R \perp A, C, B \mid E$ (Markovian property for A)
 - $R \perp A \mid E$ (using decomposition)
 - $R \perp C \mid E$ (using decomposition)
 - $R \perp B \mid E$ (using decomposition)
- Decomposition allow us to state the following
 - $X \perp \mathbf{W}$ for every $\mathbf{W} \subseteq \text{Non_Descendants}(X)$
 - Notice $\mathbf{W}$ can be any subset of $\text{Non_descendants}(X)$

$X \perp Y \cup W \mid Z$ only if
 $X \perp Y \mid Z$ and $X \perp W \mid Z$



Properties of Independence: Decomposition

- Decomposition allow us to prove the chain rule for Bayesian networks

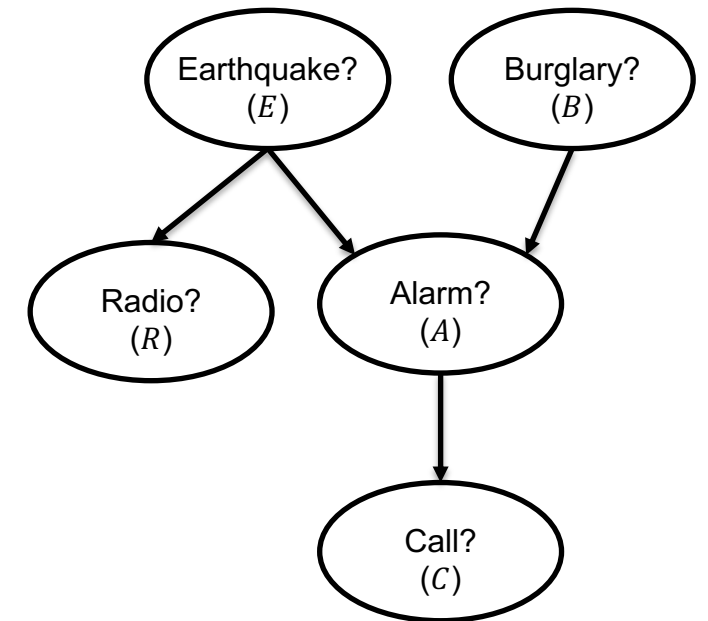
$$P(\mathbf{z}) \stackrel{\text{def}}{=} \prod_{\Theta_{x|u \sim z}} \theta_{x|u}$$

- For this example network we have

- $P(e, b, r, a, c) = \theta_e \theta_b \theta_{r|e} \theta_{a|e,b} \theta_{c|a}$
- $P(e, b, r, a, c) = P(e)P(b)P(r|e)P(a|e, b)P(c|a)$

- By the chain rule

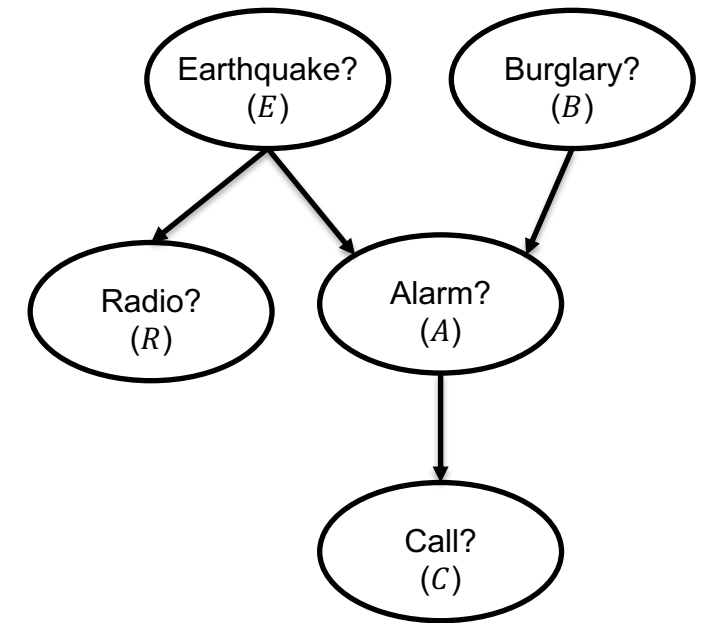
- $P(e, b, r, a, c) = P(e)P(b|e)P(r|b, e)P(a|e, b, r)P(c|a, e, b, r)$



Properties of Independence: Weak Union

- The next property is *weak union*
 - If the information $\mathbf{y}\mathbf{w}$ is not relevant to our belief in $\mathbf{x}$, then the partial information $\mathbf{y}$ will not make the rest of the information, $\mathbf{w}$, relevant
- In the example graph
 - $C \perp B, E, R \mid A$ (Markovian property for A)
 - $C \perp R \mid A, B, E$ (using decomposition)
- Weak union allow us to state the following
 - $X \perp \text{Non_Descendants}(X) \setminus \mathbf{W} \mid \text{Parents}(X) \cup \mathbf{W}$ for every $\mathbf{W} \subseteq \text{Non_Descendants}(X)$
 - This can be viewed as strengthening of the independences declared by $\text{Markov}(G)$

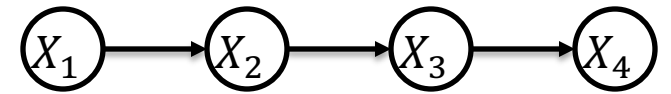
$$X \perp Y \cup W \mid Z \text{ only if } X \perp W \mid Z \cup Y$$



Properties of Independence: Contraction

- The fourth property is *contraction*
 - If after learning the irrelevant information $\mathbf{y}$ the information $\mathbf{w}$ is found to be irrelevant to our belief in $\mathbf{x}$, then the combined information $\mathbf{yw}$ must have been irrelevant from the beginning

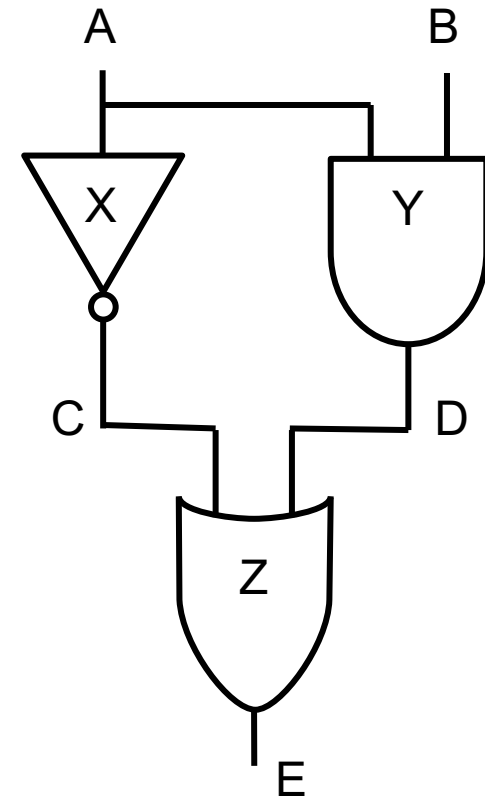
$$X \perp Y \mid Z \text{ and } X \perp W \mid Z \cup Y \text{ only if } X \perp Y \cup W \mid Z$$



Properties of Independence: Intersection

- The final axiom is *intersection*
 - It holds only for the class strictly positive distributions
 - If information w is irrelevant given y and information y is irrelevant given w , then the combined information yw is irrelevant to start with
- Symmetry, decomposition, weak union and contraction
 - Plus the property of triviality ($X \perp \emptyset \mid Z$), form the *graphoid axioms*
 - Plus intersection, the set is known as *positive graphoid axioms*

$X \perp Y \mid Z \cup W$ and $X \perp W \mid Z \cup Y$ only if
 $X \perp Y \cup W \mid Z$

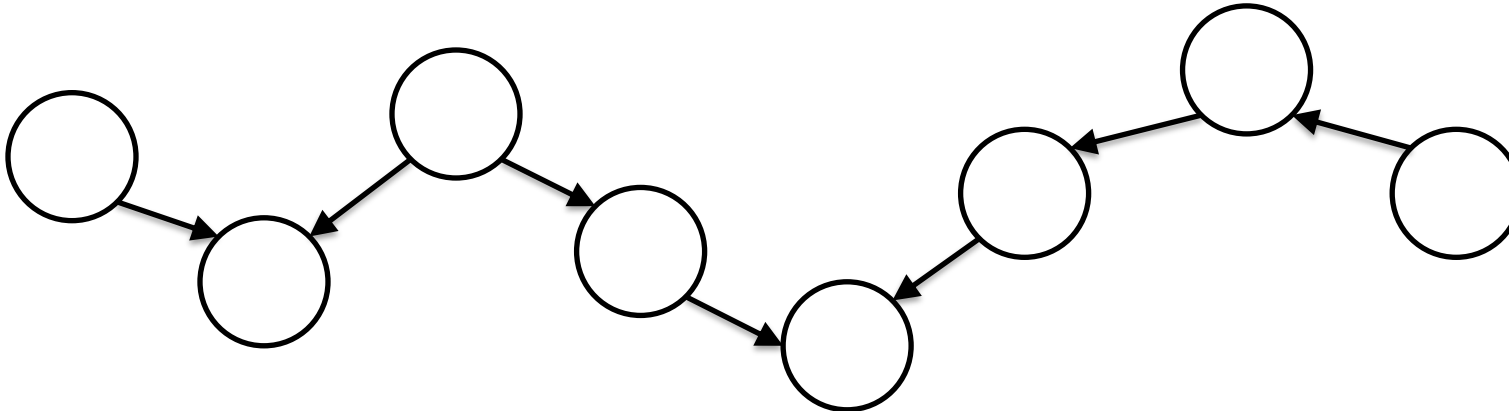
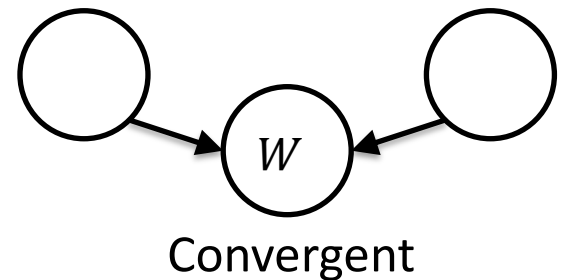
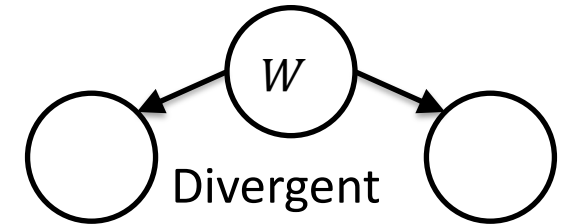
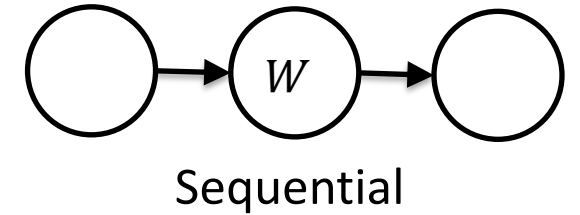


Graphical Test of Independence

- P is a distribution induced by the Bayesian network (G, Θ)
 - P satisfies independences that go beyond what is in $\text{Markov}(G)$
 - Graphoid axioms derive new independences
 - However, this derivation is not trivial
- A graphical test known as *d-separation* can capture the inferential power of graphoid axioms
 - Let X, Y , and Z be three disjoint sets of variables
 - X and Y are d-separated by Z in DAG G , if every path between a node in X and a node in Y is blocked by Z
 - If X and Y are d-separated by Z then $X \perp Y \mid Z$ for every probability distribution induced by G

Graphical Test of Independence: Blocking

- Consider this path (note that it ignores the edges direction)
 - We will view this path as a pipe and each variable W on the path as a valve
 - A valve W is either open or closed, depending on some condition
 - If at least one of the valves on the path is closed, then the whole path is blocked
 - Otherwise the path is not blocked
- There are three types of valves
 - They are determined by its relationship to its neighbours on the path



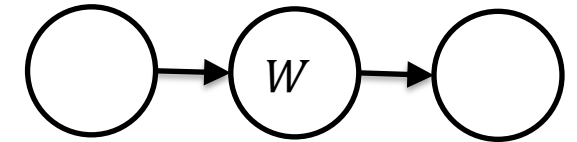
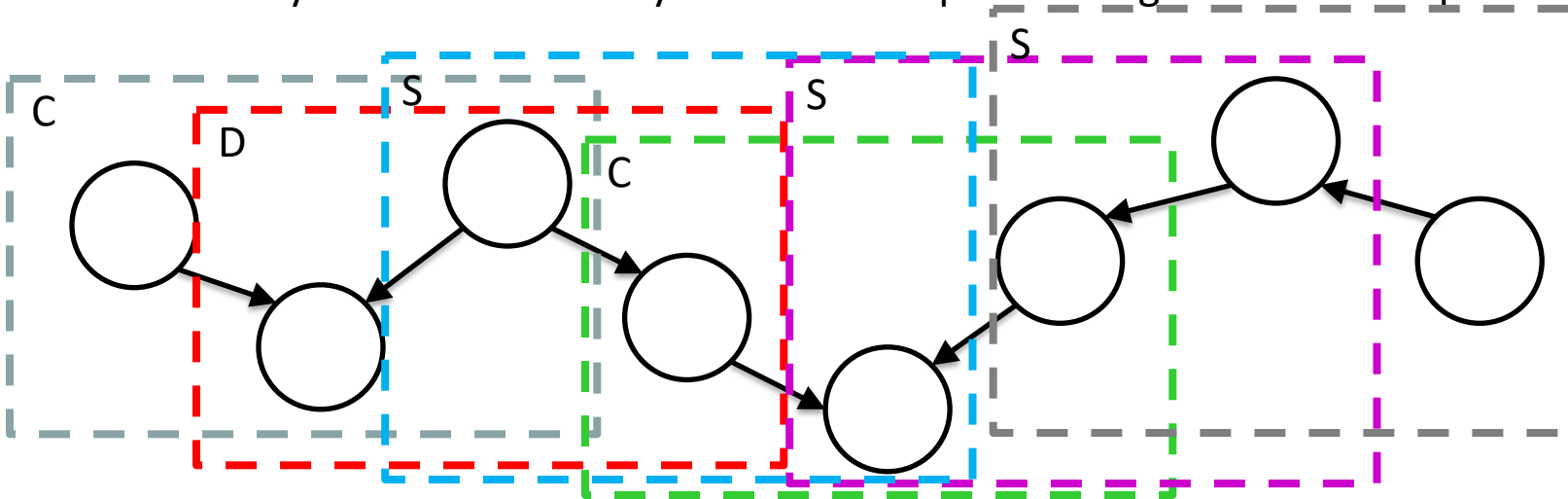
Graphical Test of Independence: Blocking

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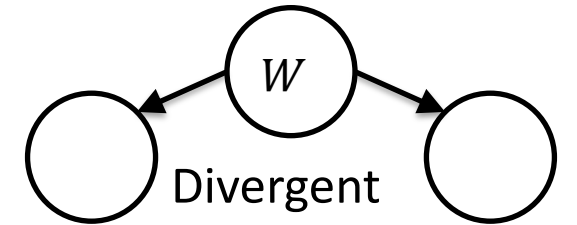
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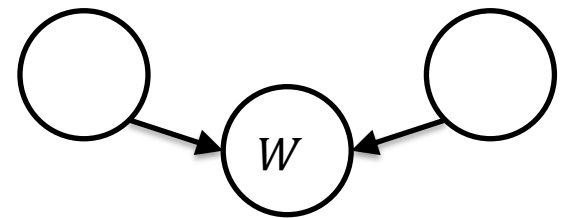
- They are determined by its relationship to its neighbours on the path



Sequential



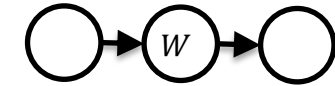
Divergent



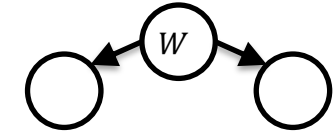
Convergent

Graphical Test of Independence: Blocking

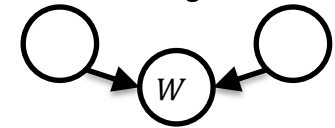
- To gain more intuition, let us use a causal interpretation
 - A sequential valve $N_1 \rightarrow W \rightarrow N_2$ declares W as an intermediary between cause N_1 and its effect N_2
 - A divergent valve $N_1 \leftarrow W \rightarrow N_2$ declares W as a common cause of two effects N_1 and N_2
 - A convergent valve $N_1 \rightarrow W \leftarrow N_2$ declares W as a common effect of two causes N_1 and N_2
- Now, we can better motivate the conditions for closed valves
 - A sequential valve is closed iff W appears in $\mathbf{Z}$
 - A divergent valve is closed iff W appears in $\mathbf{Z}$
 - A convergent valve is closed iff neither W nor any of its descendants appears in $\mathbf{Z}$



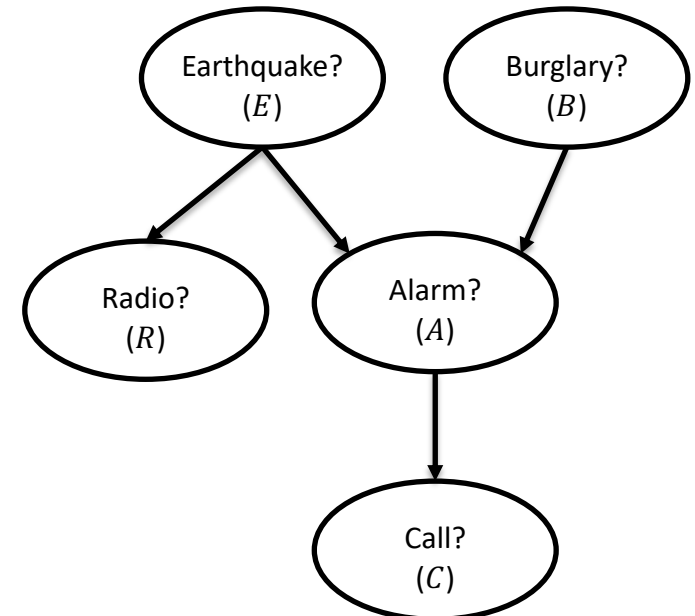
Sequential



Divergent

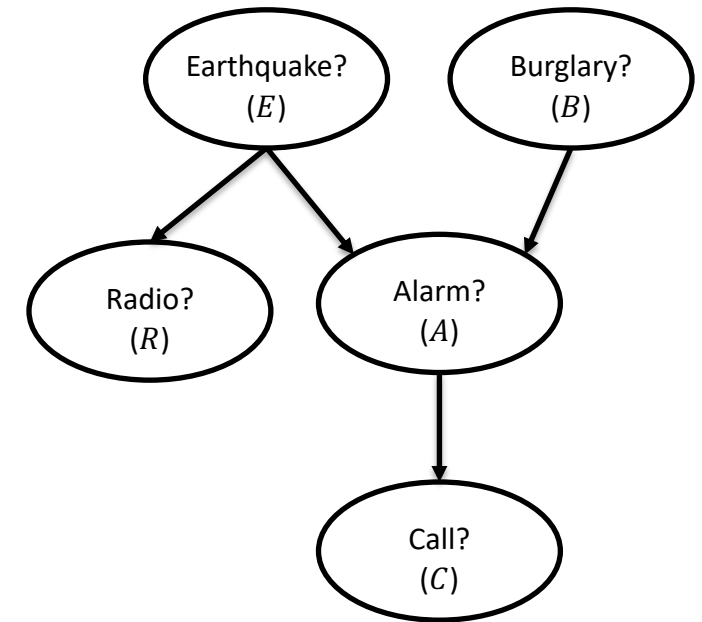


Convergent



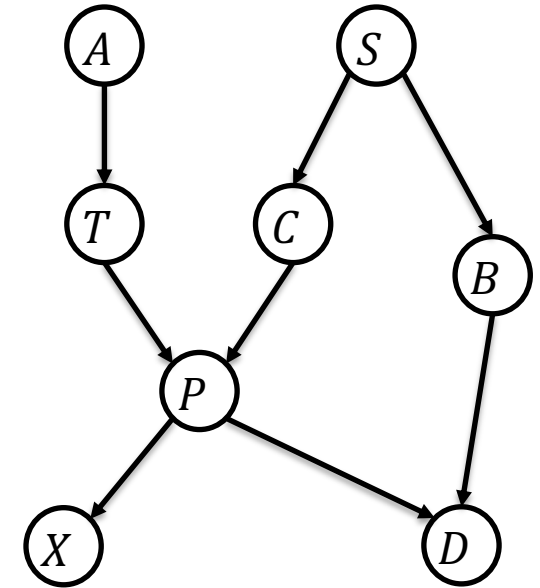
D-Separation: Definition

- Formal definition of d-separation
 - Let X , Y , and Z be disjoint sets of nodes in a DAG G . We will say that X and Y are d-separated by Z , written $dsep_G(X, Z, Y)$, iff every path between a node in X and a node in Y is blocked by Z
 - A path is blocked by Z iff at least one valve on the path is closed given Z
 - Notice that a path with no valves ($X \rightarrow Y$) is never blocked



D-Separation: Complexity

- The definition of d-separation calls for considering all paths connecting a node in X with a node in Y
 - The number of paths can be exponential
 - But we can implement a test without enumerating these paths
- Testing whether X and Y are d-separated by Z in DAG G is equivalent to testing whether X and Y are disconnected in a new DAG G' , obtained as follows
 - We delete any leaf node W from G if W does not belong to $X \cup Y \cup Z$. This process is repeated until no more nodes can be deleted
 - We delete all edges outgoing from nodes in Z
 - The connectivity test on DAG G' ignores edge direction
 - This procedure time and space are linear in the size of the DAG G



A, S d-separated from D, X by B, P ?
 T, C d-separated from B by S, X ?

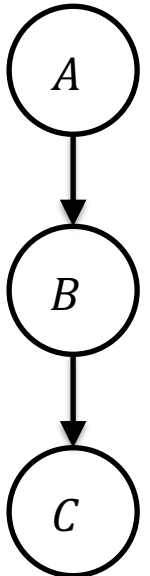
D-Separation: Soundness and Completeness

- The d-separation test is *sound*
 - If P is a probability distribution induced by a Bayesian network (G, Θ) then $dsep_G(X, Z, Y)$ only if $X \perp Y \mid Z$
 - We can safely use d-separation test to derive independence statements about the probability distributions induced by Bayesian networks
 - The proof is constructive and shows that every independence claimed by d-separation can be derived using the graphoid axioms
- The d-separation test is not *complete*
 - It is not capable of inferring every possible independence statement that holds in the induced distribution P
 - The explanation is that some independences may be hidden in the network parameters

A	θ_A
a	.6
$\bar{a}$	.4

A	B	$\theta_{B A}$
a	b	.8
a	$\bar{b}$	.2
$\bar{a}$	b	.8
$\bar{a}$	$\bar{b}$	.2

B	C	$\theta_{C B}$
b	c	.7
$\bar{b}$	$\bar{c}$	.3
b	c	.1
$\bar{b}$	$\bar{c}$	.9



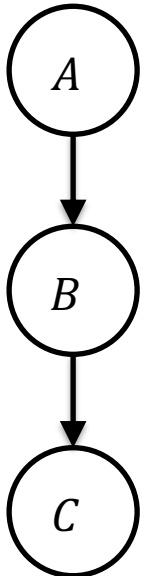
D-Separation: Soundness and Completeness

- Therefore, if we choose the parametrization carefully, we establish independences that d-separation cannot detect
 - This is not surprising since d-separation has no access to the graph parametrization
- We can conclude that, given a distribution P induced by a Bayesian network (G, Θ)
 - If X and Y are d-separated by Z , then X and Y are independent given Z for any parametrization Θ
 - If X and Y are not d-separated by Z , then whether X and Y are dependent given Z depends on the specific parametrization Θ

A	θ_A
a	.6
$\bar{a}$	.4

A	B	$\theta_{B A}$
a	b	.8
a	$\bar{b}$	.2
$\bar{a}$	b	.8
$\bar{a}$	$\bar{b}$	.2

B	C	$\theta_{C B}$
b	c	.7
$\bar{b}$	$\bar{c}$	.3
b	c	.1
$\bar{b}$	$\bar{c}$	.9



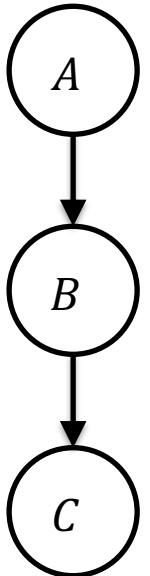
D-Separation: Soundness and Completeness

- We can always parametrize a DAG G in such a way to ensure the completeness of d-separation
- d-separation satisfies the following weak notion of completeness
 - For every DAG G , there is a parametrization Θ such that $dsep_G(\mathbf{X}, \mathbf{Z}, \mathbf{Y})$ if and only if $\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z}$
- This weaker notion of completeness implies that one cannot improve on the d-separation test
 - There is no other graphical test that can derive more independencies from G

A	θ_A
a	.6
$\bar{a}$	.4

A	B	$\theta_{B A}$
a	b	.8
a	$\bar{b}$	.2
$\bar{a}$	b	.8
$\bar{a}$	$\bar{b}$	.2

B	C	$\theta_{C B}$
b	c	.7
b	$\bar{c}$	.3
$\bar{b}$	c	.1
$\bar{b}$	$\bar{c}$	.9



Independence Maps: I-MAPs

- Independence maps describe the relationship between independence in a DAG and in a probability distribution
 - They are useful to understand the expressive power of DAGs as a language for independence statements
- Let G be a DAG and P a probability distribution over the same variables
 - G is an independence map (I-MAP) of P iff
 - It means that every independence declared by d-separation holds in P
- An I-MAP is *minimal* if G ceases to be an I-MAP if we delete any edges from G
 - If P is induced by a Bayesian network (G, Θ) , then G must be an I-MAP of P
 - But it may not be minimal

$dsep_G(X, Z, Y)$ only if $X \perp Y \mid Z$

Independence Maps: D-MAPs

- G is a dependency map (D-MAP) of P iff

$X \perp Y \mid Z$ only if $dsep_G(X, Z, Y)$

- It means that the lack of d-separation in G implies a dependence in P
- If P is induced by the Bayesian network (G, Θ) , then G is not necessarily a D-MAP of P
- G can be made a D-MAP of P if we choose the parametrization Θ carefully

Independence Maps: Perfect MAPs

- If a DAG G is both an I-MAP and a D-MAP of P , then G is a *perfect map*
 - We want G to be a P-MAP of the induced distribution to make all independences of P accessible to d-separation
 - However, there are probability distributions for which there are no P-MAPs
- Suppose we have four variables and a distribution P that *only* satisfies these dependencies
 - There is no DAG that is a P-MAP of P in this case

$$X_1 \perp X_2 \mid Y_1, Y_2$$

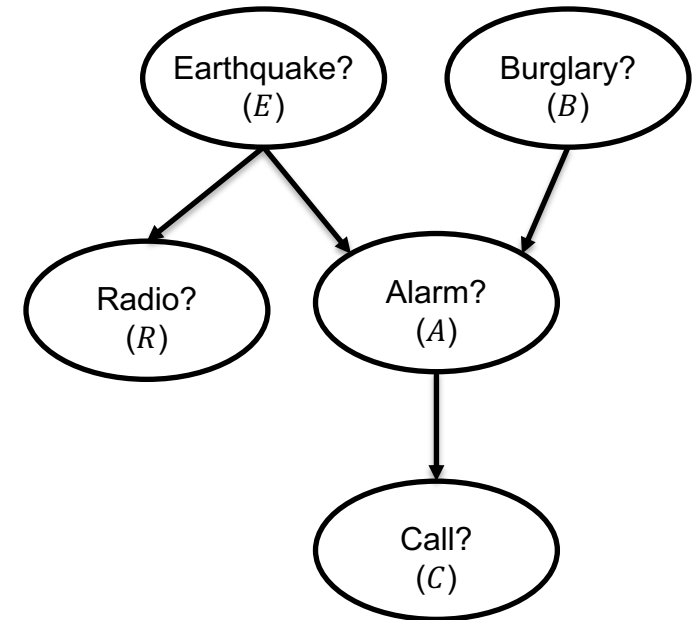
$$X_2 \perp X_1 \mid Y_1, Y_2$$

$$Y_1 \perp Y_2 \mid X_1, X_2$$

$$Y_2 \perp Y_1 \mid X_1, X_2$$

Independence Maps

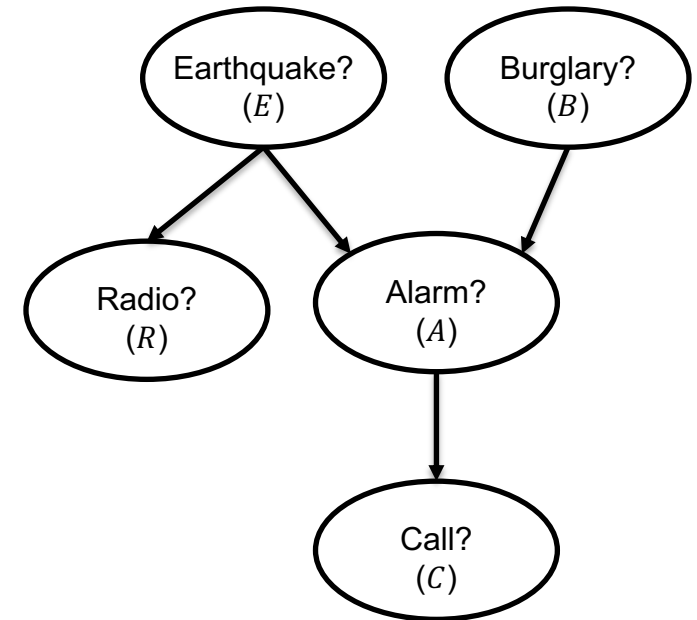
- Given a distribution P , how can we construct a DAG that is guaranteed to be a minimal I-MAP of P
 - Minimal I-MAPs tend to exhibit more independences
 - Therefore, requiring fewer parameters and leading to more compact networks
- Procedure to build a minimal I-MAP
 - Given ordering $X_1, \dots, X_n$ of variables in P
 - Start with an empty DAG G and consider the variable X_i for $i = 1 \dots n$
 - For each X_i , identify a minimal subset $\mathbf{P}$ of variables $X_1, \dots, X_{i-1}$ such that $X_i \perp X_1 \dots, X_{i-1} \setminus \mathbf{P} \mid \mathbf{P}$
 - Make $\mathbf{P}$ the parents of X_i in G



A, B, C, E, R

Independence Maps

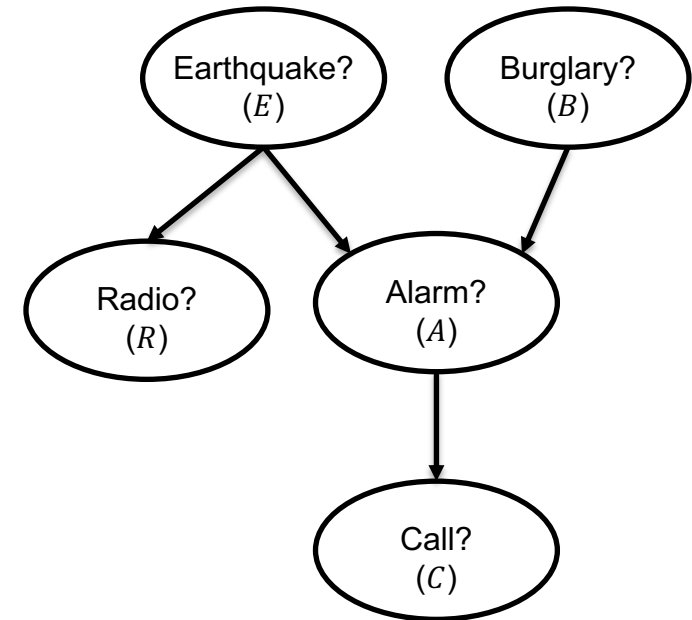
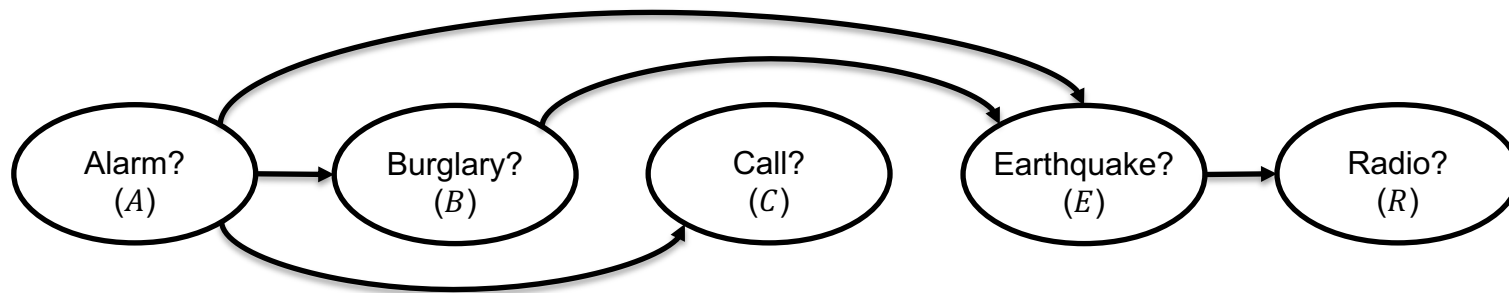
- Suppose this graph is a P-MAP of some distribution P



A, B, C, E, R

Independence Maps

- Suppose this graph is a P-MAP of some distribution P



A, B, C, E, R

$P = \emptyset$

$P = A$

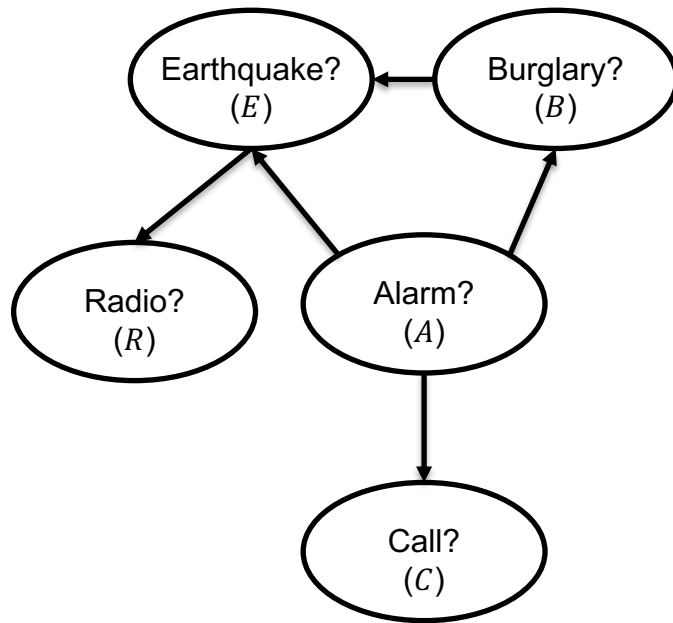
$P = A$

$P = A, B$

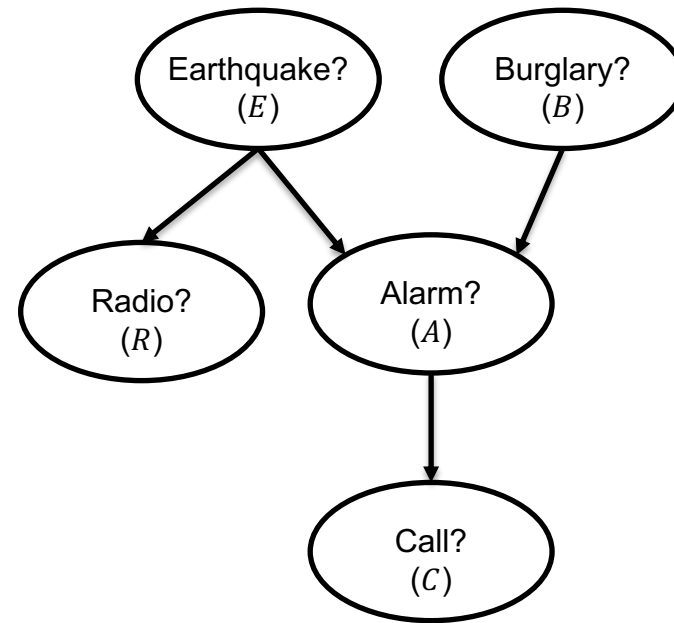
$P = E$

Independence Maps

- Suppose this graph is a P-MAP of some distribution P



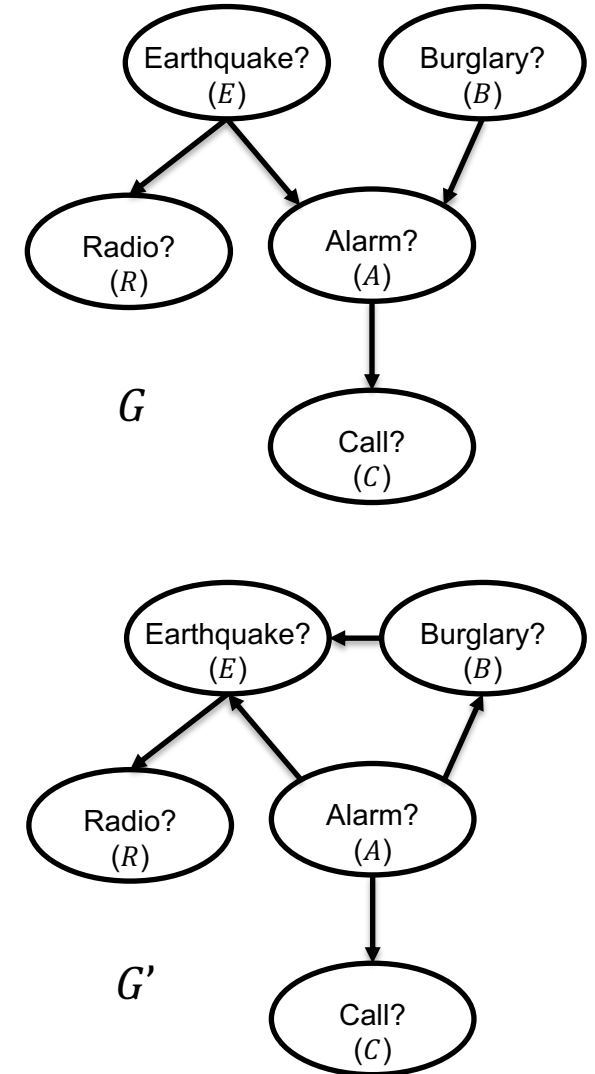
I-MAP procedure



Causal

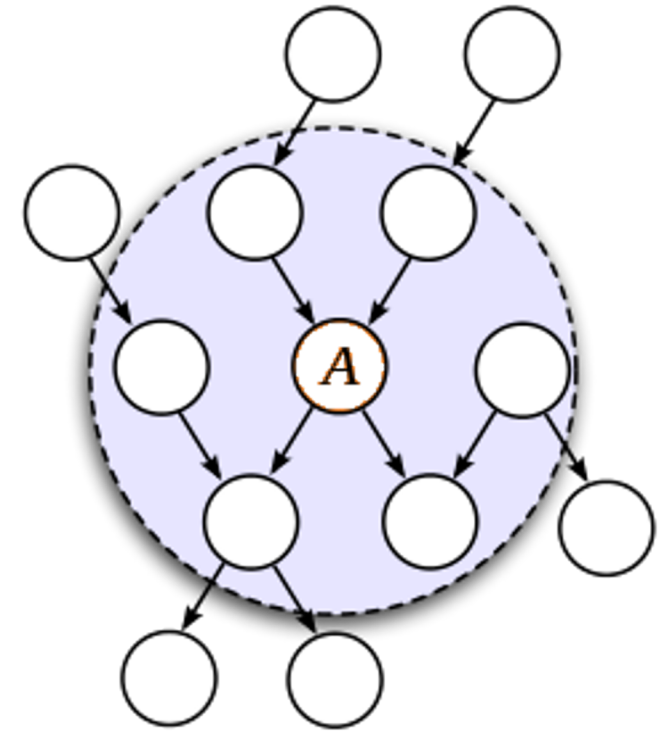
Independence Maps

- The resulting DAG G' is guaranteed to be minimal
 - d-separation in G' leads to d-separation in G and independence in P
 - This ceases to hold if we delete any edges of G'
- G' is incompatible with causal relationships
 - Yet it is sound from an independence viewpoint
 - A person that agrees with G cannot disagree with the independences in G'
- Minimal I-MAP is not unique
 - It depends of the variable ordering
 - But also we may have multiple I-MAPs for a single ordering
 - Since we may find multiple minimal sets $\mathbf{P}$ for the same variable X_i



Blankets and Boundaries

- An important notion for independence is the *Markov blanket*
 - Let P be a distribution over variables $\mathbf{X}$. A *Markov blanket* for a variable $X \in \mathbf{X}$ is the set of variables $\mathbf{B} \subseteq \mathbf{X}$ such that $X \notin \mathbf{B}$ and $X \perp \mathbf{X} \setminus (\mathbf{B} \cup \{X\}) \mid \mathbf{B}$
 - A Markov blanket for X will render every other variable irrelevant to X
 - A minimal Markov blanket is known as a *Markov boundary*. A blanket is minimal iff no strict subset of $\mathbf{B}$ is also a Markov blanket
- If P is a distribution induced by a DAG G , then a Markov blanket for X can be constructed with its parents, children, and spouses in G .
 - A variable Y is a spouse of X if the two variables have a common child in G



Conclusion

- Bayesian networks are a graphical model with a DAG
 - The graph represents the independencies between variables
 - The parametrisation expresses the strength of the dependencies
- D-separation provides a convenient and efficient approach to detect independencies
 - However, additional independencies may be hidden in the graph parametrisation
 - We also discussed the concepts of I-MAP, D-MAP and P-MAP
- Tasks
 - Read Chapter 4 from the textbook (Darwiche)